SRINIVAS UNIVERSITY

INSTITUTE OF ENGINEERING & TECHNOLOGY

Srinivas Campus,Mukka,Surathkal,Mangaluru-574 146

"RESEARCH AND PRESENT A TIMELINE SHOWING MAJOR MILESTONES IN AI HISTORY"

AN INDIVIDUAL TASK

Submitted by:

NAME	AASHITH.R.SHETTY
USN	01SU24CS004

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Prof,Mahesh Kumar

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INTRODUCTION

- Artificial Intelligence (AI) is a branch of computer science that aims to create intelligent machines capable of performing tasks that usually require human intelligence.
- AI focuses on simulating human cognitive abilities such as learning, reasoning, problem solving, perception, and decision making.
- The idea of artificial intelligence originated from the desire to understand human intelligence and replicate it using machines.
- Early AI systems worked using rule-based approaches, where machines followed predefined instructions without the ability to learn.
- Over time, limitations of rule-based systems led to the development of learning-based approaches.
- Advancements in computer hardware, increased processing power, and availability of large amounts of data contributed significantly to the growth of AI.
- Machine learning enabled systems to learn from data and improve performance without explicit programming.
- Deep learning further enhanced AI capabilities by using multi-layer neural networks to process complex data.
- Technologies such as natural language processing allow machines to understand and generate human language.
- Computer vision enables machines to interpret and analyze visual information from images and videos.
- In modern society, AI is widely used in healthcare for medical diagnosis and treatment planning.
- AI is also used in education to provide personalized learning experiences.
- In the banking and finance sector, AI helps in fraud detection and risk assessment.
- AI plays a key role in transportation through autonomous vehicles and traffic management systems.
- AI-powered tools assist software developers by automating repetitive tasks and improving productivity.
- Understanding the historical development of artificial intelligence helps in identifying major breakthroughs and challenges faced by researchers.
- A timeline of AI milestones provides a clear picture of how artificial intelligence has evolved over the years.
- Studying AI history also helps in predicting future trends and developments in the field.
- This task aims to research and present the major milestones that shaped the evolution of artificial intelligence.

DEFINITION OF ARTIFICIAL INTELLIGENCE

- Artificial Intelligence (AI) is the branch of computer science that focuses on creating machines capable of performing tasks that normally require human intelligence, such as learning, reasoning, problem solving, and decision making.
- According to **John McCarthy**, Artificial Intelligence is the science and engineering of making intelligent machines, especially intelligent computer programs.
- Artificial Intelligence enables machines to simulate human cognitive functions like thinking, understanding language, recognizing patterns, and adapting to new situations.
- AI systems are designed to analyze data, learn from experience, and improve their performance over time without constant human intervention.
- The primary goal of artificial intelligence is to develop systems that can work intelligently and efficiently in real-world environments.

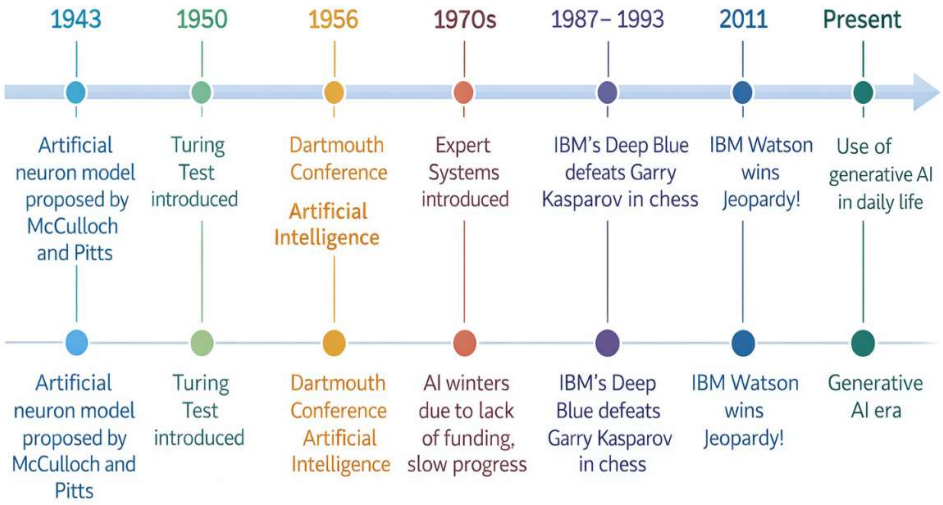
OBJECTIVES OF THE STUDY

- To understand the basic concept and meaning of Artificial Intelligence.
- To study the historical development of artificial intelligence from its origin to the present day.
- To identify the major milestones that contributed to the growth of AI.
- To analyze how artificial intelligence has evolved through different phases and eras.
- To understand the technological advancements that improved AI systems over time.
- To study the impact of artificial intelligence on various fields such as healthcare, education, banking, and transportation.
- To gain knowledge about the challenges and limitations faced during the development of AI.
- To understand the importance of artificial intelligence in modern society.
- To develop awareness about the future scope and potential of artificial intelligence.
- To present the evolution of AI in a clear and systematic timeline format.
- To understand the role of research and innovation in the advancement of artificial intelligence.
- To develop a clear understanding of how past developments in AI influence current and future technologies.

HISTORY OF ARTIFICIAL INTELLIGENCE

- 1943 – The artificial neuron model was proposed by McCulloch and Pitts, showing how machines could imitate human brain activity.
- 1950 – Alan Turing proposed the Turing Test to determine whether a machine can think like a human.
- 1956 – The term *Artificial Intelligence* was officially introduced at the Dartmouth Conference, which is considered the birth of AI research.
- 1960s – Early AI programs were developed to solve mathematical problems and play simple games.
- 1966 – ELIZA, an early chatbot, was created to simulate human conversation using pattern matching.
- 1970s – Expert systems were introduced to provide decision-making abilities in fields such as medicine and engineering.
- Late 1970s – AI systems faced limitations due to high costs and insufficient computational power.
- 1980s – Knowledge-based expert systems became popular in industrial and medical applications.
- 1987–1993 – AI research slowed down during the AI Winter due to reduced funding and unmet expectations.
- 1997 – IBM Deep Blue defeated the world chess champion, marking a major success for AI.
- 2000s – Machine learning techniques gained importance due to increased data availability and faster computers.
- 2011 – IBM Watson won the quiz show Jeopardy, demonstrating advances in natural language processing.
- 2012 – Deep learning gained attention after significant improvements in image recognition tasks.
- 2016 – AlphaGo defeated a world champion in the game of Go, a milestone in reinforcement learning.
- 2018 – AI began to be widely used in virtual assistants, recommendation systems, and autonomous vehicles.
- 2020 onwards – AI became an essential part of daily life in healthcare, education, finance, and smart devices.
- 2023 onwards – The rise of Generative AI enabled machines to create text, images, code, and videos.

Timeline Showing Major Milestones in AI History



APPLICATIONS OF ARTIFICIAL INTELLIGENCE

- Healthcare – AI is used for disease diagnosis, medical imaging, and personalized treatment plans.
- Education – AI helps in online learning platforms, automated grading, and personalized learning experiences.
- Banking and Finance – AI is used for fraud detection, credit scoring, and customer support through chatbots.
- Transportation – AI enables self-driving cars, traffic management, and route optimization.
- Retail and E-commerce – AI is used for product recommendations, demand forecasting, and customer behavior analysis.
- Manufacturing – AI helps in predictive maintenance, quality control, and automation of industrial processes.
- Entertainment – AI is used in content recommendation, gaming, and media streaming platforms.
- Agriculture – AI assists in crop monitoring, yield prediction, and smart irrigation systems.

ADVANTAGES AND DISADVANTAGES OF ARTIFICIAL INTELLIGENCE

Advantages of Artificial Intelligence	Disadvantages of Artificial Intelligence
Reduces human effort by automating repetitive tasks	High cost of development and maintenance
Works continuously without fatigue or breaks	Lack of human emotions and creativity
Provides high accuracy and reduces human error	Risk of job displacement in some sectors
Processes large amounts of data quickly	Requires large amounts of data and computing power
Improves efficiency and productivity	Dependence on machines may reduce human skills
Can perform dangerous tasks safely	Security and privacy concerns
Supports better decision-making using data analysis	Ethical and social issues related to AI usage
Enhances performance in healthcare, finance, and industry	Limited ability to handle unexpected situations

CONCLUSION

- Artificial Intelligence has grown significantly from early theoretical ideas to advanced real-world applications.
- The timeline of AI development highlights important milestones that shaped modern AI systems.
- AI is widely used in many fields such as healthcare, education, transportation, and business.
- It helps in improving efficiency, accuracy, and productivity.
- Despite its advantages, AI also has limitations like ethical issues and job displacement.
- Responsible and ethical use of artificial intelligence is essential.
- Future advancements in AI are expected to further improve human life and technology.